Lecture 5: Integration by Parts

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Perform integration by using the technique integration by parts.
- ☐ Identify the needs to use the integration by parts even the question does not suggest.
- ☐ Apply the trick of writing the integral $f(x)$ as $1 \times f(x)$ and proceed with the integration by parts.
- ☐ Solve the integration problems by using the reduction formula.

Introduction

Recall the product rule for differentiation

$$\frac{d}{dx}[u(x)v(x)] = u'(x)v(x) + u(x)v'(x).$$

What does this look like in the context of integration?

Integration by Parts

Integration by Parts Formula

$$\int u(x)v'(x)\,dx = u(x)v(x) - \int v(x)u'(x)\,dx.$$

Integration by Parts Formula - Differential Version

$$\int u\,dv = uv - \int v\,du.$$

This technique is used for the integral of the form

$$\int u(x)v(x)\,dx,$$

where one part can be easily differentiated and another part can be easily integrated.

General rule: If one of the functions in the product is a polynomial, always choose it to be u (function that get differentiated), unless the other function is $\ln x$ (or similar), in this case, choose u to be $\ln x$.

Examples

$$\int x \cos x \, dx$$

$$\int x^2 \sin x \, dx$$

Examples

$$\int \ln x \, dx$$

$$\int \arctan x \, dx$$

Examples

$$\int x^2 e^x \, dx$$

$$\int e^x \cos x \, dx$$

Examples

$$\int \cos^n x \, dx$$

$$\int e^{\sqrt{2x-1}} \, dx$$

Examples

$$\int_0^4 xe^{-x}\,dx$$

$$\int_{2\sqrt{3}}^2 x \operatorname{arcsec} x\,dx$$